MACHINE LEARNING

Ans.1 Least Square Error

Ans.2 Linear regression is sensitive to outliers

Ans.3 Negative

Ans.4 Correlation

Ans.5 Low bias and high variance

Ans.6 Predictive modal

Ans.7 Regularization

Ans.8 SMOTE

Ans.9 TPR and FPR

Ans.10 False

Ans.11 Apply PCA to project high dimensional data

Ans.12 A,B &C

Ans.13. Regularization is a technique used to reduce the errors by fitting the function appropriately on the given training set and avoid overfitting.

Regularization refers to the modifications that can be made to a learning algorithm that helps to reduce this generalization error and not the training error. It reduces by ignoring the less important features. It also helps prevent overfitting, making the model more robust and decreasing the complexity of a model.

The regularization techniques in machine learning are:

Lasso regression: having the L1 norm
Ridge regression: with the L2 norm
Elastic net regression: It is a combination of Ridge and Lasso regression.

Regularization works by shrinking the beta coefficients of a linear regression model.

Ans.14. LASSO(Least Absolute Shrinkage and Selection Operator) regression.

Ans.15. An error term represents the margin of error within a statistical model; it refers to the sum of the deviations within the regression line, which provides an explanation for the difference between the theoretical value of the model and the actual observed results.